# DSA0502 QUERY PROCESSING FOR DATASCIENCE

# CO3 – AT1 – DATA CLEANING CHALLENGE

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**QUESTION 1: E-COMMERCE CUSTOMER TRANSACTIONS DATASET**

An online retail company has collected customer transaction data from multiple sales channels.

The management wants to identify the top-performing customer segments and products.

However, the dataset contains several quality issues.

| CustomerID | CustomerName | Gender | Age | ProductCategory | Quantity | UnitPrice | PurchaseDate | City |
|---|---|---|---|---|---|---|---|---|
| C101 | Raj Kumar | Male | 28 | Electronics | 2 | 15000 INR | 10-01-2025 | Chennai |
| C102 | Priya S | Female | 35 | Fashion | 3 | 1200 INR | 11-01-2025 | chennai |
| C103 | Arun V | Male | 150 | Electronics | 1 | 300 USD | 12-01-2025 | CHENNAI |
| C104 | Kavya R | Female | 0 | Grocery | -2 | 800 INR | 15-01-2025 | Coimbatore |
| C105 | Rahul P | Male | 42 | Fashion | 5 | 2500 INR | 18-01-2025 | Madurai |
| C105 | Rahul P | Male | 42 | Fashion | 5 | 2500 INR | 18-01-2025 | Madurai |
| NULL | Sneha M | Female | 30 | Electronics | 2 | 450 USD | 05-02-2025 | Trichy |
| C107 | Deepak N | Male | | Grocery | 3 | 600 INR | 07-02-2025 | Tiruchirappalli |
| C108 | Meena K | Female | 27 | Fashion | 1 | 1800 INR | 2025-13-01 | Salem |
| C109 | John D | Male | 31 | Electronics | 4 | 250 EUR | 15-02-2025 | Chennai |

**Data Issues**

Missing Customer IDs

Duplicate transactions

Negative quantities

Unit prices recorded in different currencies

Age values such as 0, 150, and blanks

Inconsistent city names (e.g., Chennai, chennai, CHENNAI)

Incorrect purchase dates

**Tasks**

1. Identify and classify all data quality issues. (5 Marks)

2. Design a cleaning strategy for each issue. (5 Marks)

3. Perform data standardization and transformation. (5 Marks)

4. Analyze how cleaning affects total revenue calculations. (5 Marks)

5. Generate a report discussing business risks if the data were not cleaned. (5 Marks)

**SOLUTION:**

**1. IDENTIFY AND CLASSIFY ALL DATA QUALITY ISSUES**

The dataset contains the following quality problems:

| Issue | Example | Classification |
|---|---|---|
| Missing Customer ID | NULL for Sneha M | Completeness |
| Duplicate transaction | C105 appears twice | Duplication |
| Negative quantity | C104 has -2 | Validity |
| Different currencies | INR, USD, EUR | Consistency |
| Invalid age | Age 0, 150 | Validity |
| Missing age | C107 has no age | Completeness |
| Inconsistent city names | Chennai, chennai, CHENNAI | Consistency |
| Incorrect date | 2025-13-01 | Validity |
| Inconsistent date formats | 10-01-2025 and 2025-13-01 | Consistency/Validity |
| Missing/unclear values | C107 has missing Age and some formatting issues | Completeness |
| Currency included with price | 15000 INR, 300 USD | Formatting/Consistency |

**Explanation**

- **Completeness:** Some required fields such as Customer ID and Age are missing.
- **Validity:** Age 0/150, negative quantity, and invalid dates are not acceptable values.
- **Consistency:** City names and currency formats are inconsistent.
- **Uniqueness:** C105 is repeated, causing duplicate revenue.
- **Accuracy:** Currency differences can make revenue calculations incorrect if not converted.

**2. CLEANING STRATEGY FOR EACH ISSUE**

**a) Missing Customer ID**

The NULL Customer ID should be investigated using other transaction information.

- If the customer can be uniquely identified, assign the correct ID.
- If it cannot be identified, mark it as Unknown rather than inventing an ID.

**b) Duplicate Transactions**

The two identical C105 records should be checked.

Since all transaction details are identical, one record should be removed.

**Before:** 10 records
**After removing duplicate:** 9 records

**c) Negative Quantity**

C104 has quantity -2.

- Check the original transaction.
- If it represents a return, record it separately as a return/refund.
- If it is an error, correct it using the original transaction record.

For normal sales analysis, the invalid -2 should not be treated as a normal sale.

**d) Different Currencies**

Prices are recorded in INR, USD and EUR.

All prices should be converted into a **single currency**, preferably INR.

For example:

Revenue = Quantity × Standardized Unit Price

Current exchange rates should be used when performing the actual conversion.

**e) Invalid Age**

Age values such as 0 and 150 are unrealistic.

- Replace invalid ages with NULL.
- If the correct age is available from another reliable source, use that value.
- Do not replace it with an arbitrary value.

**f) Missing Age**

For missing age:

- Check the customer database for the correct age.
- If unavailable, retain it as NULL.
- For customer segmentation, exclude unknown ages or place them in an "Unknown" category.

**g) Inconsistent City Names**

Standardize city names using proper capitalization.

For example:

- Chennai
- chennai
- CHENNAI

should all become:

**Chennai**

Similarly, Tiruchirappalli should be used consistently instead of Tiruchirappalli with line breaks.

**h) Incorrect Purchase Date**

2025-13-01 is invalid because there is no 13th month.

The original transaction should be checked to determine the correct date.

It should **not** be automatically changed without verification.

### 3. DATA STANDARDIZATION AND TRANSFORMATION

After cleaning, the dataset should follow a common format.

| Customer ID | Customer Name | Gender | Age | Category | Quantity | Unit Price | Currency | Purchase Date | City |
|---|---|---|---|---|---|---|---|---|---|
| C101 | Raj Kumar | Male | 28 | Electronics | 2 | 15000 | INR | 10-01-2025 | Chennai |
| C102 | Priya S | Female | 35 | Fashion | 3 | 1200 | INR | 11-01-2025 | Chennai |
| C103 | Arun V | Male | NULL/Verified | Electronics | 1 | 300 | USD | 12-01-2025 | Chennai |
| C104 | Kavya R | Female | NULL/Verified | Grocery | Invalid | 800 | INR | 15-01-2025 | Coimbatore |
| C105 | Rahul P | Male | 42 | Fashion | 5 | 2500 | INR | 18-01-2025 | Madurai |
| C106/Unknown | Sneha M | Female | 30 | Electronics | 2 | 450 | USD | 05-02-2025 | Trichy |
| C107 | Deepak N | Male | NULL | Grocery | 3 | 600 | INR | 07-02-2025 | Tiruchirappalli |
| C108 | Meena K | Female | 27 | Fashion | 1 | 1800 | INR | **Invalid—verify** | Salem |
| C109 | John D | Male | 31 | Electronics | 4 | 250 | EUR | 15-02-2025 | Chennai |

**Transformations**

1. **Customer ID:** Standard format such as C101, C102, etc.
2. **Gender:** Convert Femal e → Female.
3. **Age:** Store only realistic values.

4. **Quantity:** Store as numeric values and reject invalid negative quantities.
5. **Unit Price:** Separate numeric price from currency.
6. **Currency:** Convert all prices to INR.
7. **Date:** Convert all dates to one format such as YYYY-MM-DD.
8. **City:** Standardize spelling and capitalization.
9. **Revenue:** Create a calculated field:

**Revenue = Quantity × Unit Price**

After currency conversion:

**Revenue (INR) = Quantity × Unit Price (INR)**

## 4. EFFECT OF CLEANING ON TOTAL REVENUE

Data cleaning has a major effect on revenue calculations.

Consider the valid INR transactions first:

**C101**

2 × ₹15,000 = ₹30,000

**C102**

3 × ₹1,200 = ₹3,600

**C105**

5 × ₹2,500 = ₹12,500

There are **two C105 records**, so if duplicates are not removed:

₹12,500 + ₹12,500 = ₹25,000

But the correct revenue from C105 is only:

₹12,500

**C107**

3 × ₹600 = ₹1,800

Therefore, the valid INR revenue from these transactions is:

**₹30,000 + ₹3,600 + ₹12,500 + ₹1,800 = ₹47,900**

The USD and EUR transactions cannot be directly added to this amount because they must first be converted to INR.

**Effect of Data Problems**

| Problem | Effect on Revenue |
|---|---|
| Duplicate C105 | **Overestimates** revenue |
| Negative quantity | May produce incorrect/negative revenue |
| USD/EUR values | Incorrect totals if added directly to INR |

| | |
|---|---|
| Missing Customer ID | Revenue cannot be properly linked to customer |
| Invalid dates | Revenue may be assigned to the wrong period |
| Incorrect prices | Revenue becomes inaccurate |

**Conclusion**

Without cleaning, the company could report an **incorrect total revenue**. Duplicate records may inflate revenue, while invalid quantities and inconsistent currencies can either inflate or reduce the calculated amount.

## 5. BUSINESS RISKS IF DATA IS NOT CLEANED

Poor-quality data can create several business risks.

**1. Incorrect Revenue Reporting**

Duplicate transactions can make sales appear higher than they actually are.

**2. Wrong Customer Segmentation**

Missing or incorrect age and customer IDs can result in incorrect customer groups.

For example, management may incorrectly identify a particular age group as the most profitable.

**3. Incorrect Product Performance**

Invalid quantities and prices can make some products appear more or less profitable than they really are.

**4. Financial Loss**

Currency differences can cause incorrect revenue calculations and financial reporting.

**5. Poor Business Decisions**

Managers may invest more money in products or cities that only appear successful because of bad data.

**6. Incorrect Sales Forecasting**

Invalid dates can affect monthly or yearly sales trends, leading to poor forecasting.

**7. Customer Management Problems**

Missing customer IDs make it difficult to track customer purchases and identify valuable customers.

**8. Loss of Trust**

Incorrect reports can reduce confidence in the company's analytics and decision-making systems.

## OVERALL CONCLUSION

Data cleaning is essential before performing e-commerce analysis. The dataset should first be checked for **missing values, duplicates, invalid values, inconsistent formats, currency**

**differences, and incorrect dates**. After standardization, revenue can be calculated accurately and management can reliably identify **top-performing customers, products, categories, and cities**.

**In short:**
**Raw Data → Cleaning → Standardization → Transformation → Accurate Revenue Analysis → Better Business Decisions**

**QUESTION 5: GLOBAL SALES PERFORMANCE DATASET**

**1. Comprehensive Data-Cleaning Framework for Multinational Datasets**

A systematic data-cleaning framework can be applied in the following stages:

**Step 1: Data Profiling**

First, inspect the dataset for:

- Missing values
- Duplicate records
- Invalid ratings
- Different currencies
- Inconsistent dates
- Different product names
- Salesperson spelling variations
- Extreme sales values

**Step 2: Remove Duplicates**

The Germany transaction for **Anna M – Laptop – €1200** appears twice. One duplicate record should be removed after verification.

**Step 3: Validate Data**

Check whether:

- Customer ratings are between **1 and 5**
- Sales amounts are positive
- Dates are valid
- Currency codes are valid
- Product and salesperson names follow a standard format

**Step 4: Standardize Data**

Convert currencies, dates, product names, salesperson names, and region names into common formats.

**Step 5: Handle Missing and Abnormal Values**

Missing sales amounts should be investigated and extreme values should be checked against the original source.

**Step 6: Quality Validation**

After cleaning, perform another quality check to ensure that the dataset contains no remaining duplicates, invalid values, or inconsistent formats.

**2. Methods to Standardize Currencies, Products and Dates**

**A. Currency Standardization**

The dataset contains **INR, USD, EUR and GBP**.

All currencies should be converted into one base currency, preferably **USD** or **INR**.

For example:

**Sales in base currency = Original Sales Amount × Exchange Rate**

Exchange rates should be taken from a reliable source corresponding to the transaction date.

| Currency | Action |
| --- | --- |
| INR | Convert to base currency |
| USD | Convert to base currency |
| EUR | Convert to base currency |
| GBP | Convert to base currency |

This prevents incorrect comparisons between countries.

**B. Product Standardization**

Products are recorded in different languages.

For example:

- Laptop
- Ordinateur Portable

Both should be mapped to a common product name such as:

**Laptop**

A product master/reference table can be used:

| Original Name | Standard Name |
| --- | --- |
| Laptop | Laptop |
| Ordinateur Portable | Laptop |
| Mobile | Mobile |
| Smartphone | Smartphone |

If **Mobile** and **Smartphone** represent the same product in the company's product catalog, they can also be mapped to a single standardized category.

**C. Date Standardization**

Different date formats are used:

- 10-01-2025
- 12-01-2025
- 01-15-2025

All dates should be converted to one standard format such as:

**YYYY-MM-DD**

For example:

- 10-01-2025 → 2025-01-10
- 12-01-2025 → 2025-01-12
- 01-15-2025 → 2025-01-15

The date format should be interpreted according to the country/source system before conversion to avoid confusing **DD-MM-YYYY** with **MM-DD-YYYY**.

**3. Handling Missing Values and Outliers**

**A. Missing Sales Amount**

The Priya S Mobile transaction has:

**SalesAmount = NULL**

The following methods can be used:

1. Check the original sales system or invoice.
2. If unavailable, estimate using appropriate methods such as:
    - Median sales for the same product
    - Median sales for the same country/region
    - Historical sales for the same customer/product
3. If reliable estimation is impossible, retain it as missing rather than inventing a value.

For financial reporting, **verification from the original source is preferred**.

**B. Invalid Customer Ratings**

The valid rating range is **1–5**.

Problematic values:

- USA → Rating **6**
- India → Rating **0**

These values should be treated as invalid.

The records should be checked against the original customer feedback system. If the correct rating cannot be found, replace it with NULL rather than assuming a rating.

### C. Extreme Sales Outlier

The India transaction:

**Rahul P – Mobile – ₹9,999,999**

is extremely high compared with the other transactions.

Possible techniques for detecting outliers include:

**IQR Method**

Calculate:

**IQR = Q3 − Q1**

Values below:

**Q1 − 1.5 × IQR**

or above:

**Q3 + 1.5 × IQR**

can be treated as potential outliers.

**Z-Score Method**

A value with a very large absolute Z-score, commonly above **3**, can be investigated as an outlier.

However, an outlier should **not automatically be deleted**. The original invoice should be checked first.

If ₹9,999,999 is a data-entry error, it should be corrected. If it is a genuine large transaction, it should be retained.

## 4. Regional Sales Rankings Before and After Cleaning

### Before Cleaning

The raw dataset gives the following approximate regional totals, without correcting currencies, duplicates, missing values, or the extreme outlier:

| Rank | Region | Raw Sales |
|---|---|---|
| 1 | Asia | ₹10,084,999 |
| 2 | Europe | €2,500 |
| 3 | North America | $1,500 |
| 4 | UK | £1,100 |

**Important:** These rankings are **not meaningful** because different currencies are being added and compared directly.

Asia also appears extremely strong because of the **₹9,999,999 outlier**.

**After Cleaning**

After cleaning:

- Remove the duplicate Germany transaction.
- Convert all currencies into a common base currency.
- Correct/remove the erroneous extreme value if verified as an error.
- Handle the missing sales amount.
- Standardize regions and product names.

The regional ranking should then be recalculated using:

**Regional Sales = Sum of standardized sales amounts**

The exact final ranking depends on the **exchange rates used** and whether the ₹9,999,999 transaction is confirmed as genuine or erroneous.

**Expected analytical effect**

| Issue | Effect on Ranking |
|---|---|
| Currency differences | Can create false regional comparisons |
| Duplicate Germany record | Artificially increases Europe |
| ₹9,999,999 outlier | Artificially increases Asia |
| Missing sales amount | Underestimates Asia |
| Standardization | Makes regional comparison meaningful |

Therefore, the **raw ranking should not be used for management decisions**.

**5. Executive Report: Impact of Data Quality on Strategic Decisions**

**Executive Summary**

The multinational sales dataset contains several data-quality problems, including inconsistent currencies, missing sales amounts, duplicate records, invalid customer ratings, inconsistent dates, multilingual product names, salesperson spelling variations, and extreme sales values.

These problems can significantly affect management's understanding of regional and product performance.

**Key Business Impacts**

**1. Incorrect Regional Performance**

If currencies are not converted into a common currency, management cannot accurately compare countries or regions.

**2. Incorrect Revenue Forecasting**

Extreme or duplicate sales values can make historical sales appear unusually high, resulting in inaccurate future forecasts.

**3. Poor Product Decisions**

Different names for the same product can cause sales to be split across multiple categories. This may make a popular product appear less successful.

**4. Incorrect Employee Performance**

Raj Kumar and Rajkumar may represent the same salesperson. Without standardization, their sales could be counted separately.

**5. Incorrect Customer Analysis**

Ratings such as 0 and 6 can distort customer satisfaction analysis.

**6. Financial Risk**

Missing and incorrect sales amounts can lead to inaccurate revenue reports and financial planning.

**7. Strategic Risk**

Management may incorrectly allocate marketing budgets, inventory, employees, and investments to regions that only appear successful because of poor-quality data.

**Final Conclusion**

A multinational company must clean and standardize its data before calculating regional performance or forecasting future sales.

The recommended process is:

**Data Profiling → Validation → Duplicate Removal → Missing Value Handling → Outlier Detection → Currency Conversion → Product Standardization → Date Standardization → Regional Analysis → Forecasting**

Clean data provides **reliable sales rankings, accurate forecasts, better resource allocation, and more confident strategic decisions**.